STAGE 3: DEPLOYMENT & APP STORE

Complete Deployment Guide - App Store, Payments, Costs

Project: [YOUR PROJECT NAME]
Timeline: Day 5 (8 hours)
Goal: Live in production on all platforms

SECTION 1: PRE-DEPLOYMENT CHECKLIST

1.1 Final Verification

- [ ] All tests passing
- [ ] No console errors
- [ ] No console warnings
- [ ] Performance acceptable
- [ ] Security verified
- [ ] Database backed up
- [ ] Rollback plan ready

1.2 Deployment Checklist

Code Quality

- [ ] Code reviewed
- [ ] No hardcoded credentials
- [ ] No console.log statements
- [ ] No TODO comments left
- [ ] All imports used
- [ ] No dead code

Security

- [ ] API keys in environment variables
- [ ] Database password secure
- [ ] HTTPS enabled
- [ ] Security headers configured
- [ ] CORS configured correctly
- [ ] Rate limiting enabled

Performance

- [ ] Page load time <3s
- [ ] API response time <500ms
- [ ] Bundle size <500KB
- [ ] Images optimized
- [ ] Caching configured

Data

- [ ] Database migrated
- [ ] Database backed up

- [ ] Test data removed
- [ ] Production data ready

SECTION 2: WEB APP DEPLOYMENT

2.1 Build Production Bundle

Step 1: Build

```
npm run build
# Output: dist/ folder
```

Step 2: Verify Build

```
# Check for errors
npm run build 2>&1 | grep -i error

# Check bundle size
du -sh dist/
# Should be <500KB
```

Step 3: Test Production Build

```
npm run preview
# Visit http://localhost:4173
# Test all features
```

2.2 Deploy to Vercel (Recommended)

Step 1: Connect Repository

```
npm install -g vercel
vercel login
vercel link
```

Step 2: Configure Environment

```
# Set environment variables
vercel env add DATABASE_URL
vercel env add STRIPE_SECRET_KEY
vercel env add STRIPE_PUBLIC_KEY
# ... add all other secrets
```

Step 3: Deploy

```
vercel deploy --prod
```

Step 4: Verify

- [ ] App loads
- [ ] All features work
- [ ] No errors
- [ ] Performance acceptable

2.3 Deploy to Other Platforms

Render:

```
# Connect GitHub
# Select repository
# Deploy
```

AWS:

```
# Build
npm run build

# Upload to S3
aws s3 sync dist/ s3://your-bucket/

# Invalidate CloudFront
aws cloudfront create-invalidation --distribution-id YOUR_ID --paths "/*"
```

Cost: $0-50/month (depending on traffic)

SECTION 3: iOS APP DEPLOYMENT

3.1 Prerequisites

- [ ] Apple Developer account ($99/year)
- [ ] Xcode installed
- [ ] Provisioning profiles created
- [ ] Signing certificates created

3.2 Build for iOS

Step 1: Build Web

```
npm run build
```

Step 2: Copy to iOS

```
npx cap copy ios
```

Step 3: Open Xcode

```
npx cap open ios
```

Step 4: Configure App

- [ ] Set app name
- [ ] Set bundle ID (com.yourcompany.app)
- [ ] Set version (1.0.0)
- [ ] Set build number (1)
- [ ] Add app icon
- [ ] Add launch screen

Step 5: Build Archive

- Select “Generic iOS Device”
- Product → Archive
- Wait for build to complete

Step 6: Validate

- Select archive
- Click “Validate App”
- Fix any issues

3.3 Submit to App Store

Step 1: Create App Store Listing

- Open App Store Connect
- Click “My Apps”
- Click “+”
- Create new app
- Fill in app information

Step 2: Fill App Information

- [ ] App name: [Your app name]
- [ ] Subtitle: [Short description]
- [ ] Description: [Full description]
- [ ] Keywords: [Relevant keywords]
- [ ] Support URL: [Your support URL]
- [ ] Privacy Policy URL: [Your privacy policy]

Step 3: Add Screenshots

- [ ] iPhone 6.5” (1284x2778)
- [ ] iPhone 5.5” (1242x2208)
- [ ] iPad 12.9” (2048x2732)
- [ ] Show key features in screenshots

Step 4: Set Pricing

- [ ] Select pricing tier
- [ ] Set availability date
- [ ] Select countries

Step 5: Upload Build

- Select archive
- Click “Distribute App”
- Select “App Store Connect”
- Upload build

Step 6: Submit for Review

- Fill in review information
  - [ ] Contact information
  - [ ] Demo account (if needed)
  - [ ] Notes for reviewer
- Click “Submit for Review”

Step 7: Wait for Approval

- Typically 24-48 hours
- Check email for updates
- Fix any rejection issues

3.4 Common iOS Rejection Reasons

Rejection 1: "Misleading App Description"

- **Fix**: Ensure description matches functionality
- **Prevent**: Test app against description

Rejection 2: "Crashes on Real Devices"

- **Fix**: Test on real device before submission
- **Prevent**: Use TestFlight for beta testing

Rejection 3: "External Payment Methods"

- **Fix**: Use In-App Purchase for digital goods
- **Prevent**: Use Stripe/Razorpay for physical goods only

Rejection 4: "Hardcoded Credentials"

- **Fix**: Move API keys to environment variables
- **Prevent**: Scan code before submission

Rejection 5: "Privacy Policy Missing"

- **Fix**: Add privacy policy link in app
- **Prevent**: Include in app settings

SECTION 4: ANDROID APP DEPLOYMENT

4.1 Prerequisites

- [ ] Google Play Developer account ($25 one-time)
- [ ] Android Studio installed
- [ ] Signing key created
- [ ] Keystore file created

4.2 Build for Android

Step 1: Build Web

```
npm run build
```

Step 2: Copy to Android

```
npx cap copy android
```

Step 3: Open Android Studio

```
npx cap open android
```

Step 4: Create Signing Key

```
# In Android Studio:
# Build → Generate Signed Bundle/APK
# Select "Bundle (Google Play)"
# Create new keystore
# Fill in key information
# Remember password!
```

Step 5: Build Bundle

- Select “Build Bundle (Google Play)”
- Select signing key
- Wait for build

Step 6: Verify

- Check for errors
- Check bundle size
- Check permissions

4.3 Submit to Google Play

Step 1: Create Play Store Listing

- Open Google Play Console
- Click “Create app”
- Fill in app name
- Select category

Step 2: Fill App Information

- [ ] App name: [Your app name]
- [ ] Short description: [Short description]
- [ ] Full description: [Full description]
- [ ] Screenshots: [At least 2]
- [ ] Feature graphic: [1024x500]

- [ ] Icon: [512x512]
- [ ] Privacy policy: [URL]

Step 3: Add Screenshots

- [ ] Phone (1080x1920)
- [ ] Tablet (1440x2560)
- [ ] Show key features

Step 4: Set Pricing

- [ ] Select pricing tier
- [ ] Select countries
- [ ] Set release date

Step 5: Upload Bundle

- Select bundle file
- Upload
- Wait for processing

Step 6: Submit for Review

- Fill in review information
  - [ ] Contact information
  - [ ] Demo account (if needed)
  - [ ] Notes for reviewer
- Click “Submit”

Step 7: Wait for Approval

- Typically 2-4 hours
- Check email for updates
- Fix any rejection issues

4.4 Common Android Rejection Reasons

Rejection 1: "Misleading App Description"

- **Fix**: Match description to functionality

Rejection 2: "Excessive Permissions"

- **Fix**: Only request necessary permissions
- **Prevent**: Explain why each permission needed

Rejection 3: "Crashes on Real Devices"

- **Fix**: Test on real device

Rejection 4: "Inappropriate Content"

- **Fix**: Ensure content rating accurate

Rejection 5: "Malware Detected"

- **Fix**: Scan for malware before submission

SECTION 5: PAYMENT GATEWAY SETUP

5.1 Stripe Setup

Step 1: Create Account

- Visit https://stripe.com
- Click "Sign up"
- Enter email and password
- Verify email

Step 2: Get API Keys

- Go to Dashboard

- Click “Developers”
- Click “API keys”
- Copy Publishable Key and Secret Key

Step 3: Add to Environment

```
# .env
STRIPE_PUBLIC_KEY=pk_test_...
STRIPE_SECRET_KEY=sk_test_...
```

Step 4: Configure Webhook

- Go to Developers → Webhooks
- Click “Add endpoint”
- Enter URL: https://yourapp.com/webhook/stripe
- Select events: payment_intent.succeeded, payment_intent.payment_failed
- Copy signing secret

Step 5: Add Webhook Secret

```
# .env
STRIPE_WEBHOOK_SECRET=whsec_...
```

Step 6: Test Payment

- Use test card: 4242 4242 4242 4242
- Verify payment succeeds
- Verify webhook fires
- Verify invoice marked as paid

5.2 Razorpay Setup

Step 1: Create Account

- Visit https://razorpay.com
- Click “Sign up”
- Enter email and password
- Verify email

Step 2: Get API Keys

- Go to Settings → API Keys
- Copy Key ID and Key Secret

Step 3: Add to Environment

```
# .env
RAZORPAY_KEY_ID=...
RAZORPAY_KEY_SECRET=...
```

Step 4: Configure Webhook

- Go to Settings → Webhooks
- Click “Add new webhook”
- Enter URL: https://yourapp.com/webhook/razorpay
- Select events: payment.authorized, payment.failed
- Copy webhook secret

Step 5: Add Webhook Secret

```
# .env
RAZORPAY_WEBHOOK_SECRET=...
```

Step 6: Test Payment

- Use test UPI: success@razorpay
- Verify payment succeeds
- Verify webhook fires
- Verify invoice marked as paid

SECTION 6: COST BREAKDOWN

6.1 Monthly Infrastructure Costs

Service	Cost	Notes
Frontend Hosting (Vercel)	$0-20	Free tier for MVP
Backend Hosting (Render)	$7-50	$7 starter, 50 production
Database (PostgreSQL)	$15-100	$15 for 1GB, 100 for 10GB
CDN (Cloudflare)	$0-20	Free tier or $20/month
Email (SendGrid)	$0-10	Free 100/day or $10/month
Storage (AWS S3)	$0-10	$0.023 per GB
Monitoring (Sentry)	$0-29	Free tier or $29/month
Total	**$22-239**	**Depends on scale**

6.2 One-Time Costs

Service	Cost	Notes
Domain name	$10-15	Namecheap, GoDaddy
SSL certificate	$0	Free with Let's Encrypt
iOS Developer	$99	Annual fee
Android Developer	$25	One-time fee
Total	**$134-139**	**One-time**

6.3 Payment Processing Costs

Provider	Fee	Notes
Stripe	2.9% + $0.30	Per transaction
Razorpay	2% + $0.30	Per transaction (India)
Example		
$100 payment (Stripe)	$3.20	
$100 payment (Razorpay)	$2.30	

6.4 Total First Month Cost

Category	Cost
Infrastructure	$50-100
One-time setup	$134-139
Payment processing	2-3% of revenue
Total	**$184-239 + payment fees**

6.5 Monthly Recurring Cost

Category	Cost
Infrastructure	$50-100
Payment processing	2-3% of revenue
Total	**$50-100 + payment fees**

SECTION 7: DEPLOYMENT CHECKLIST

7.1 Pre-Deployment (Day 5, 8 AM)

- [ ] All tests passing
- [ ] No console errors
- [ ] Database backed up
- [ ] Rollback plan ready
- [ ] Monitoring configured
- [ ] Alerts configured

7.2 Web Deployment (Day 5, 9 AM)

- [ ] Build production bundle
- [ ] Test production build
- [ ] Deploy to staging
- [ ] Test on staging
- [ ] Deploy to production
- [ ] Verify production
- [ ] Monitor for errors

7.3 iOS Deployment (Day 5, 11 AM)

- [ ] Build archive
- [ ] Validate archive
- [ ] Create App Store listing
- [ ] Upload build
- [ ] Submit for review
- [ ] Wait for approval

7.4 Android Deployment (Day 5, 12 PM)

- [ ] Build bundle
- [ ] Create Play Store listing
- [ ] Upload bundle
- [ ] Submit for review
- [ ] Wait for approval

7.5 Post-Deployment (Day 5, 2 PM)

- [ ] Monitor error rates
- [ ] Monitor performance
- [ ] Monitor uptime
- [ ] Collect user feedback
- [ ] Fix critical issues
- [ ] Plan next features

SECTION 8: MONITORING & SUPPORT

8.1 Setup Monitoring

Error Tracking:

```
npm install @sentry/react
# Configure Sentry
```

Performance Monitoring:

```
# Use Vercel Analytics
# Automatic for Vercel deployments
```

Uptime Monitoring:

```
# Use UptimeRobot
# Monitor https://yourapp.com every 5 minutes
```

8.2 Setup Alerts

Email Alerts:

- [ ] Error rate >1%
- [ ] Response time >500ms
- [ ] Uptime <99.9%
- [ ] Payment failures

Slack Alerts:

- [ ] Critical errors
- [ ] Payment failures
- [ ] Deployment failures

8.3 Support Process

User Support:

- [ ] Email support: support@yourapp.com
- [ ] Response time: <24 hours
- [ ] Resolution time: <48 hours

Bug Reports:

- [ ] Collect error details

- [ ] Reproduce bug
- [ ] Fix and deploy
- [ ] Notify user

CONCLUSION

Deployment stage complete when:

- ✅ Web app live
- ✅ iOS app submitted
- ✅ Android app submitted
- ✅ Payment gateway working
- ✅ Monitoring active
- ✅ Support ready

Next Step: Move to STAGE 4 - PRODUCTION PDF

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